

Harpuneet Singh

Toronto, ON | 437-335-7184 | harpuneet.singh@torontomu.ca | github.com/fadeddart | LinkedIn | Portfolio

EDUCATION

Toronto Metropolitan University

Toronto, ON

Bachelor of Science in Computer Science (Co-op)

Sep 2024 – May 2028

– **CGPA:** 3.90 | **Dean's List:** Fall 2024, Winter 2025, Fall 2025, Winter 2026

– **Coursework:** Data Structures & Algorithms, Database Systems, Object-Oriented Design, Software Engineering

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, C/C++, HTML/CSS

Frameworks & Libraries: FastAPI, SQLAlchemy, asyncio, Node.js, PostgreSQL

Tools & Platforms: Docker, Git, GitHub Actions, Linux, REST APIs, Prometheus

Concepts: Distributed Systems Basics, Concurrency, CI/CD, Testing, Agile, System Design

EXPERIENCE

Student Developer

May 2026 – Present

Toronto Metropolitan University

Toronto, ON

– Shipped internal web applications for university staff workflows, turning stakeholder requirements into production UI and application logic

– Built and iterated JavaScript/HTML/CSS features through Agile cycles with testing, debugging, and code review

– Improved maintainability with reusable front-end structure so follow-on changes required less rework

Accounting Assistant

Jan 2026 – May 2026

HNM Logistics

Mississauga, ON

– Automated multi-thousand-row reporting workflows with Python scripts, replacing fragile manual spreadsheet steps

– Implemented validation checks that flagged operational/financial inconsistencies before close

FEATURED TECHNICAL PROJECTS

FORGE — DAG Workflow Engine | *Python, FastAPI, asyncio, Docker,*

GitHub

Prometheus

– Designed a concurrent DAG workflow engine with durable task state and a bounded worker pool, sustaining **7,000+** task completions/min under load tests

– Implemented dependency-aware scheduling, idempotent submission, exponential backoff retries, dead-letter isolation, and stuck-run recovery

– Exposed Prometheus metrics plus `/health/ready`; packaged with Docker Compose for one-command local reproduction

ATLAS — RAG Knowledge System + Eval Harness | *Python, FastAPI, BM25, TF-IDF*

GitHub

– Built an end-to-end RAG platform with hybrid BM25 + TF-IDF retrieval, extractive cited answers, and a **54**-question evaluation harness

– Raised retrieval doc-precision from **0.52** (BM25-only) to **0.62** (hybrid); **47/54** eval cases passed with **0.92+** faithfulness

– Added query caching with TTL and latency telemetry; documented Azure OpenAI / AI Search deployment path

AEGIS — Fuzzing & Vulnerability Discovery Harness | *Python, C, Docker*

GitHub

– Engineered a mutation-based fuzzing harness against custom C/Python parsers with crash triage, deduplication, and minimized PoC inputs

– Documented **3** unique lab defect classes (buffer overflow, null deref, logic/recursion) with reproducible crash artifacts